

Sai Vijay Ragav

+91 93458 93521 | saivijayragav@gmail.com | linkedin.com/in/sai-vijay-ragav | github.com/saivijayragav

EDUCATION

Rajalakshmi Institute of Technology

B.E. in Computer Science (Artificial Intelligence & Machine Learning)

Chennai, Tamil Nadu

2023 – 2027

Indian Institute of Technology

Diploma in Data Science

Chennai, Tamil Nadu

2024 – 2026

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Dart

Tech: Spring Boot, FastAPI, REST APIs, Microservices, LangChain, LangGraph

Databases/Tools: PostgreSQL, Firebase, Milvus, Git, Docker, Maven

Core: Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems, Computer Networks

EXPERIENCE

AI/ML Intern

Oct 2025 – Dec 2025

Spirelia

Remote

- Developed a multimodal RAG pipeline using **LangChain**, **Unstructured**, and **Milvus** to index **10,000+ patent documents** and support semantic retrieval across **1M+ embeddings** with sub-second query latency.
- Enriched documents with metadata to improve downstream retrieval and knowledge discovery.

Software Development Intern

May 2025 – Aug 2025

National Informatics Centre

Chennai, Tamil Nadu

- Built a **Spring Boot** ticketing platform integrated with **FastAPI**, **Qwen2.5-VL**, and PostgreSQL, automating extraction of **8+ metadata fields** and reducing ticket submission effort by **60–70%**.
- Developed a **RAG-based** support assistant spanning **10+ ticket categories** across **20+ states**.

PROJECTS

BinarySync (*GitHub*) | Java, Spring Boot, AWS S3

May 2026 – Jun 2026

- Built a cloud-native file versioning system using **content-defined chunking (CDC)** and **SHA-256 deduplication** to transfer only modified portions of large binary files.
- Implemented concurrent S3 transfers using **Java 21 Virtual Threads** and manifest-based reconstruction through Spring Boot REST APIs.

Printing Shop Management App (*GitHub*) | Java, Spring Boot, Flutter, Cloudflare R2

Oct 2025 – Jan 2026

- Built a Flutter application with **Spring Boot REST APIs** for document uploads, order processing, and real-time tracking across a **5,000+ student** campus environment.
- Integrated **Razorpay**, **Firebase**, and **Cloudflare R2** into a scalable workflow supporting high-volume uploads, automated payments, and notifications.

NeuroScreen (*GitHub*) | Python, FastAPI, Flutter, Machine Learning

Feb 2026 – Jul 2026

- Built a multimodal dementia screening platform using **FastAPI** and Flutter, integrating speech, physical activity, and caregiver assessments.
- Implemented confidence scoring, longitudinal progress tracking, and AI-driven therapy recommendations.

ACHIEVEMENTS

Hackathons (2026): 1st Place — BNY Service Design Jam, AI Wars, Tekhora'26 | **2nd Place** — VYUHATHON 1.0 | **3rd Place** — DEV ARENA'26 | India Innovates 2026 Participant (*Drive*)

Competitive Programming: Solved **600+** LeetCode problems (**Knight Badge**, Max Rating: 1968); secured **Global Rank 1749** in **TCS CodeVita Season 13**.

AI Azure Internship Program (Microsoft Initiative) (May–Jun 2025): Built an ML-based AI vs Human text detection system during the AICTE–Edunet program. (*Drive*)

AWS Academy Cloud Foundations Graduate (Apr 2024): Hands-on experience with core AWS services and cloud fundamentals. (*Drive*)